

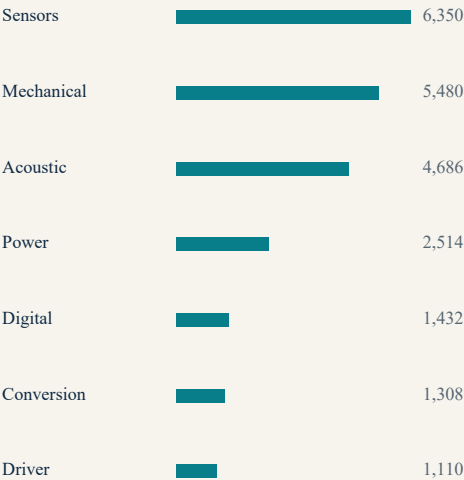
PRELIMINARY - NOT ORDER AUTHORITY

INR 26,148

quantity-1 component and unit-material estimate; excludes laboratory instruments and labour

Subsystem	Qty-1 INR	Share	Procurement note
Sensors	6,350	24.3%	MS5837 pressure sensor is long-lead and not in hand.
Mechanical	5,480	21.0%	Wet penetrators and V2 pod are required for water release.
Acoustic	4,686	17.9%	Finished-band PZT is the largest technical and quotation risk.
Power	2,514	9.6%	Standard planning estimate; verify before purchase.
Digital	1,432	5.5%	Standard planning estimate; verify before purchase.
Conversion	1,308	5.0%	Standard planning estimate; verify before purchase.
Driver	1,110	4.2%	Standard planning estimate; verify before purchase.
PCB	800	3.1%	4-layer price includes a landed planning allowance, not a live cart quote.
Analog	750	2.9%	Standard planning estimate; verify before purchase.
Interconnect	600	2.3%	Standard planning estimate; verify before purchase.
Telemetry	570	2.2%	Standard planning estimate; verify before purchase.
Assembly	500	1.9%	Standard planning estimate; verify before purchase.
Protection	48	0.2%	Standard planning estimate; verify before purchase.

Cost concentration



Evidence boundary: DAA screenshots identify Robu.in SKUs and quantities for the bench parts. Current captured web prices include ESP32-S3 DevKitC-1 N8 at INR 1,282, MCP4921T-E/SN at INR 388 and GU1008C-40R at INR 186. Every other value is an INR planning estimate pending a dated quotation.

SeaNergy Bill of Materials

One line per item | Every status and substitute field populated

ID	Subsystem	Refs	MPN	Qty	Unit INR	Ext INR	Lead	Status	Substitute
B001	Power	F1	MF-MSMF200X-2	1	45	45	7 d	needed	Littelfuse 2016L200
B002	Power	D1	SMBJ15A	1	35	35	7 d	needed	SMBJ14A
B003	Power	Q1	DMP3010LK3-13	1	55	55	7 d	needed	AO3401A
B004	Power	U1	LM2596S-ADJ-MODULE	1	180	180	5 d	in hand	MP1584 module
B005	Power	U2	AP2112K-3.3TRG1	1	45	45	7 d	needed	TLV75533PDBVR
B006	Power	L1-L3	BLM31PG601SN1L	3	18	54	7 d	needed	MPZ3216S601A
B007	Power	C1-C12	GRM32ER71E106KA12L	12	25	300	7 d	in hand	CL32B106KAJNNNE
B008	Digital	U3	ESP32-S3-DevKitC-1-N8	1	1,282	1,282	30 d	needed	Waveshare ESP32-S3-N8R8
B009	Digital	D2	USBLC6-2SC6	1	60	60	7 d	needed	TPD2EUSB30
B010	Digital	J1	USB4105-GF-A	1	90	90	14 d	needed	TYPE-C-16PIN
B011	Conversion	U4	MCP4921T-E/SN	1	388	388	7 d	in hand	MCP4921-E/P
B012	Conversion	RN1-RN2	NOMCA16031002AT1	2	350	700	21 d	needed	LT5400 custom ratio
B013	Conversion	U5	SN74LVC16373DLR	1	220	220	14 d	needed	74ALVC16373
B014	Analog	U6	OPA1652AIDR	1	540	540	14 d	needed	OPA1642AIDR
B015	Analog	U6B-DNP	MCP6002-E/P	1	210	210	5 d	in hand	not populated for 200 kHz
B016	Driver	U7	TC4427AVOA	1	210	210	14 d	needed	MCP14A0602
B017	Driver	Q2-Q5	AO3400A	4	25	100	7 d	needed	IRLML6344
B018	Driver	T1	SEANERGY-TX-01	1	800	800	42 d	needed	air-core matching network
B019	Acoustic	BZ1	GU1008C-40T/R	1	186	186	5 d	in hand	HY40A12T/R
B020	Acoustic	BZ2	PZT-200K-WB	1	4,500	4,500	56 d	needed	switched narrowband PZT bank
B021	Sensors	S1	DS18B20-PROBE	1	220	220	5 d	in hand	PT100 plus MAX31865
B022	Sensors	S2	TDS-MODULE-V1	1	850	850	7 d	in hand	SEN0244
B023	Sensors	S3	TURBIDITY-MODULE	1	1,100	1,100	7 d	in hand	SEN0189
B024	Sensors	S4	MS5837-30BA	1	3,800	3,800	35 d	needed	Keller 7LD
B025	Sensors	U8	ADS1115IDGSR	1	380	380	14 d	needed	ADS1015IDGSR
B026	Telemetry	U9	INA226AIDGSR	1	320	320	14 d	needed	INA219BIDCNR
B027	Telemetry	DS1	SSD1306-OLED-096	1	250	250	5 d	in hand	SH1106 module
B028	Protection	D3-D6	BAT54S	4	12	48	7 d	needed	BAV99
B029	Interconnect	J2-J7	43045-0200	6	100	600	14 d	needed	JST-VH equivalent
B030	PCB	PCB1	SEANERGY-REV-A	1	800	800	14 d	needed	PCB Power India quote
B031	Mechanical	ENC1	SEANERGY-POD-V2	1	1,200	1,200	7 d	needed	SLS PA12 production pod
B032	Mechanical	OR1-OR2	AS568-246-NBR70	2	250	500	14 d	needed	FKM 75 AS568-246
B033	Mechanical	PEN1-PEN2	WETLINK-M10	2	1,800	3,600	30 d	needed	SubConn Micro Circular
B034	Power	BAT1	3S-2600-LIPO	1	1,800	1,800	14 d	needed	3S Li-ion protected pack
B035	Mechanical	H1-H12	M3X12-A4-70	12	15	180	5 d	needed	316 SS M3 x 12
B036	Assembly	MAT1	SEANERGY-CONSUMABLES	1	500	500	5 d	needed	equivalent qualified kit

The production MPN for BZ2 remains a selected placeholder, so fabrication release is blocked until a vendor-characterised part is approved.